# ANUSHKA JHA *she/her*

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## Summary

AI/ML Engineer pursuing a Master's in Computer Science at Arizona State University with hands-on experience developing LLM applications, computer vision models, and end-to-end machine learning pipelines. Skilled in Python, PyTorch, Transformers, and Retrieval-Augmented Generation (RAG), with a passion for building practical AI solutions that solve real-world problems.

## Education

**Arizona State University** — ***Tempe, Arizona***
*Master of Science in Computer Science* — ***Aug 2025 – May 2027***
**SRM Institute of Science and Technology** — ***India***
*Bachelor of Technology in Computer Science and Engineering (AI and ML specialization)* — ***Aug 2021– May 2025***
**Relevant Coursework:** Artificial Intelligence, Machine Learning, Deep Learning, Data Mining, Data Structures & Algorithms, Database Management Systems, Probability & Statistics, Linear Algebra

## Technical Skills

**Programming** : Python, C++, C
**AI / Machine learning :** PyTorch, TensorFlow, Deep Learning, Machine Learning, Transformers, XGBoost, LLMs, Natural Language Processing, Computer Vision, Prompt Engineering, Structured Outputs, Retrieval-Augmented Generation (RAG)
**DataBase** : SQL, Relational Database, Data processing, Batch processing
**Tools :** Git, Linux, VS Code, Codex, Claude Code, Cursor

## Work Experience

***Machine Learning Intern*** — ***Nov 2024 – May 2025***
***Fxis.AI***

- Built a Python pipeline that extracted structured candidate information from resumes using the OpenAI API, reducing manual resume screening effort and achieving approximately 85% extraction accuracy through manual evaluation.
- Designed prompts and validation logic to extract information consistently from resumes with different layouts while preventing missing information from being filled with incorrect or inferred data.
- Developed a recommendation system that ranked relevant job content using SQL queries and ranking logic, increasing recommendation engagement by approximately 20–25%.
- Automated candidate communication by integrating the Twilio WhatsApp API with the recruitment workflow, maintaining reliable message delivery for hiring campaigns.

***Machine Learning Intern*** — ***May 2024 – Jun 2024***
***Prodigy InfoTech***

- Developed a convolutional neural network (CNN) for hand gesture recognition using PyTorch and OpenCV, contributed to a vision system that classified hand gestures with over 90% test accuracy.
- Prepared and preprocessed image datasets by resizing, normalizing, and augmenting training images, improving model robustness across different lighting conditions and hand positions.
- Worked closely with the development team to integrate the trained model into the overall gesture recognition application and validate its performance on real image samples.

## Projects

***Gait Recognition and Classification using Deep Learning*** | *Research Paper (Publication in review)*

- Built an end-to-end deep learning pipeline on CASIA-B (124 subjects; front, side and back views), reaching **95.54%** walking-condition accuracy and **91.33%** subject-identification accuracy.
- Fused pretrained CNN features (ResNet101 + EfficientNetB2, VGG16 + InceptionV3), reduced them to 512 dimensions with PCA, and benchmarked 18 configurations of temporal, convolutional and residual models, including LSTM, BiLSTM, CNN-LSTM, 3D CNN and GaitNet.
- Cleaned the dataset by removing about 1.8% of corrupted or empty silhouettes. Improved robustness with augmentation, class weighting, early stopping and learning-rate scheduling.
- Evaluated each model with precision, recall, F1-score, confusion matrices and per-angle accuracy across separate pipelines for walking condition and person identification.

***Various Machine Learning Model for Water Quality Classification***| *Published Research Paper*

- Developed an end-to-end machine learning pipeline to classify water potability using physicochemical features, supporting automated water quality assessment for environmental monitoring.
- Prepared and analyzed the dataset through exploratory data analysis (EDA), mean imputation for missing values, outlier handling, feature scaling, and train-test splitting to improve data quality before model training.
- Evaluated Support Vector Machine (SVM), Decision Tree, and XGBoost models using grid search and performance metrics including accuracy, precision, recall, F1-score, and confusion matrices, with XGBoost achieving the best classification accuracy of 93.07%.
- Analyzed feature relationships and model behavior to identify key indicators of water quality, improving model interpretability and supporting data-driven model selection.